GRIVIN ROERING

Industrial Designer | Product Development, Design for Manufacture, CNC & Fabrication
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Portfolio: gtroeringdesign.work | linkedin.com/in/gtroeringdesign
Code B driver's licence, own vehicle | English (first language), Afrikaans (basic)

PROFESSIONAL SUMMARY

Industrial design engineer with a BSc from The Hague University of Applied Sciences, a BCom in Marketing and Management Sciences, and 10 years of hands-on manufacturing experience across steel fabrication, CNC machining and product development. Sole designer on the largest bouldering gym in Africa (approximately 900m², Fourways, Johannesburg), taking the facility from concept design through every CNC production file to completed installation. Combines CAD and design-for-manufacture capability with workshop-level knowledge of materials, supplier coordination and on-site project delivery.

TECHNICAL SKILLS

CAD and 3D Modelling: Fusion 360 (advanced), Gravity Sketch (intermediate), SolidWorks (basic, working towards certification), Shapr3D, SketchUp, Blender and Rhino (basic)

Manufacturing and Prototyping: CNC router programming in Fusion CAM (advanced), 3D printing and slicing (advanced), CNC machining, design for manufacture (DFM), steel fabrication and welding, blacksmithing, woodwork, rapid prototyping

Project Delivery: Technical drawing packs, bills of materials (BOM), material sourcing and supplier negotiation, production planning, contractor supervision and training, budgeting and scheduling

Commercial: Client and account management, quoting, team management, business administration

WORK EXPERIENCE

Lead Designer – Bloc 11, Fourways, Johannesburg January 2025 – July 2025

- Sole designer on the design and construction of an approximately 900m² bouldering gym, the largest in Africa, from concept through to opening.
- Designed the complete climbing wall system and the steel superstructure supporting it.
- Produced every CNC cutting file for the wall panels and trained a machine operator, bringing panel production in-house.
- Sourced suppliers and coordinated all material and equipment orders with the business owner.
- Supervised steel fabrication and on-site installation of the climbing walls, monitoring quality and progress against schedule.
- Produced architectural-style layout drawings used for the construction of gym amenities, and contributed to final venue selection.
- Designed an adjustable-angle Kilterboard training wall driven by an acme-screw linear actuator for the Cape Town branch.

Head of Manufacturing and Maintenance – Bloc 11, Cape Town January 2021 – July 2022

- Designed and manufactured furniture, shop fittings and custom climbing volumes for two bouldering facilities, and kept all workshop machinery and gym equipment operational.
- Developed an in-house volume texturing method that brought products up to competition standard and removed reliance on external suppliers.
- Introduced lightweight plywood construction for large climbing volumes, reducing handling weight and making installation easier for route setters.
- Managed workshop scheduling, shared-space tenants and external CNC and manufacturing commissions.
- Completed the BCom in Marketing and Management Sciences alongside this role.

Steel Design and Construction Contractor – Bloc 11 Gym Build, Cape Town January 2020 – December 2020

- Designed the structural steel superstructure for the climbing walls and devised the method of suspending the walls from it.
- Pioneered a suspended-panel system in which CNC-cut timber panels assembled onto the steel frame using connector arms, reducing steel cost, assembly complexity and build time.
- Converted 3D models into CNC-cuttable files, planned material requirements, issued purchase orders and delegated

work to the construction team.

Founder – Struk Iron, Cape Town November 2018 – December 2019

- Founded and ran a steel fabrication and blacksmithing workshop serving private and corporate clients, managing design, quoting, production, branding and one employee.
- Designed, manufactured and installed a suspended ceiling for the Swish Property Group offices, the largest commission undertaken by the business.

Apprentice Blacksmith and Co-Manager – The Harbour Forge, Cape Town August 2016 – November 2018

- Trained in blacksmithing and steel fabrication under a master smith; designed, quoted and forged bespoke ironwork including an organic wrought-iron garden gate.
- Restructured the administrative side of the business and managed selected clients and their projects.

SELECTED ACADEMIC PROJECT

Election Chair for the Municipality of The Hague – The Hague University of Applied Sciences 2025 – 2026

- Managed all of the CAD and laser cutting detailing on 2m high sculptural hands. Further managed sourcing a supplier that could deliver the project within budget, created comprehensive cutting lists, developed full assembly instructions, and devised a system of assembly and disassembly.
- Constructed from treated pine and marine plywood. Unveiled by the Mayor of The Hague in early 2026 leading up to the municipal elections.

EDUCATION AND QUALIFICATIONS

BSc Industrial Design Engineering – The Hague University of Applied Sciences, Netherlands 2023 – 2026

Pathway Programme: Mathematics and Physics – The Hague University of Applied Sciences 2023

BCom Marketing and Management Sciences – IMM Graduate School 2019 – 2021

Digital Marketing (online short course) – University of Cape Town 2017

National Senior Certificate – Calibre Education 2015

ADDITIONAL INFORMATION

Portfolio: gtroeringdesign.work

Reference: Marijus Šmigelskis, Founder, Bloc 11 | +27 71 808 5848 | marijus3@gmail.com

Written reference letter available on request.

Interests: Rock climbing and bouldering